



**Pediatric Emergency  
Research Network**



# Core Principles for Managing Acute Pain and Distress in Children

**Itai Shavit, MD**

Pain relief as a  
human right

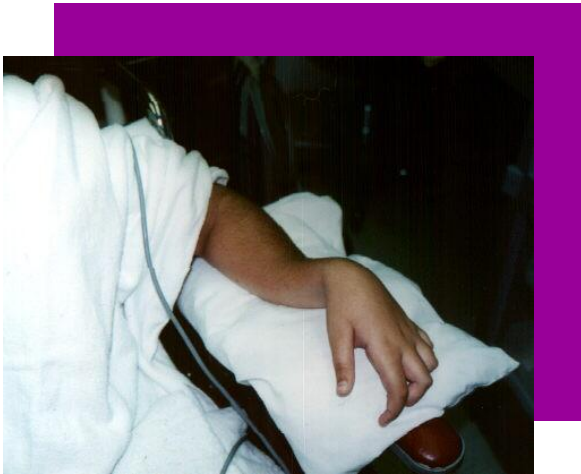
## Pain relief is an implied universal human right

**“The unreasonable failure to treat pain is poor medicine, unethical practice, and is an abrogation of a fundamental human right”**

Joint Commission on Accreditation of Healthcare Organizations *Comprehensive Accreditation Manual for Hospitals* Oakbrook Terrace, IL . Joint Commission on Accreditation of Healthcare Organizations 2001

Brennan F, Carr DB, Cousins M. Pain management: a fundamental human right. *Anesth Analg*. 2007;105(1):205-21.

# Acute Pain and suffering



**Procedural pain and distress**

# Children are different



**Procedural pain and distress**

# **Anxiety around strangers**



# **Fear / Distress**



# **Procedural pain and distress**

“is this baby suffering” ?  
(pain/distress)



**Procedural pain and distress**

# Neonatal Facial Coding System (NFCS)



Pain in young infants can be accurately assessed



The NFCS is a valid and reliable tool

brow bulge, eye squeeze, naso-labial furrow, horizontal mouth stretch, vertical mouth stretch , open lips, taut tongue

**Procedural pain and distress**



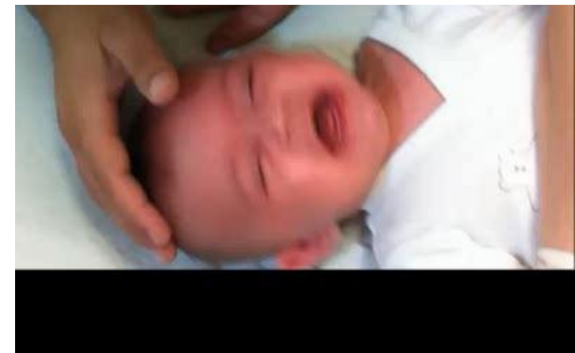
# Discomfort

Inserting a medical instrument into a body orifice can cause significant discomfort



≠

# Distress







# **CULTURAL BARRIERS**

**Procedural pain and distress**

# The Paradox of Culture

**Procedural pain and distress**

**Don't do to children what you  
would not want done to you**



Why doing it this way ?

**Procedural pain and distress**



Ashura  
ritual  
(Shia)

India



Lebanon

**Procedural pain and distress**



Kenya



Sudan

**Procedural pain and distress**



Significant advancements have occurred in the field of acute pediatric pain management over the past two decades

**Procedural sedation and analgesia**





The NEW ENGLAND  
JOURNAL of MEDICINE

**SOUNDING BOARD**

# Pain, Hurt, and Harm -- The Ethics of Pain Control in Infants and Children

Gary A. Walco, Ph.D., Robert C. Cassidy, Ph.D., and Neil L. Schechter, M.D.

N Engl J Med 1994; 331:541-544 | August 25, 1994 | DOI: 10.1056/NEJM199408253310812

"...The assessment and treatment of pain in children are important parts of Pediatric practice, and failure to provide adequate control of pain amounts to substandard and unethical medical practice

**Procedural pain and distress**

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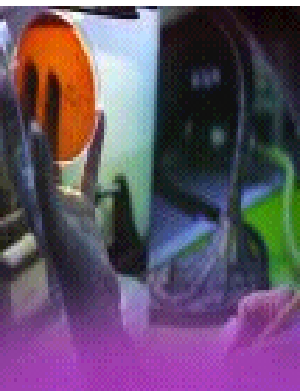
AMERICAN ACADEMY OF PEDIATRICS

# The Assessment and Management of Acute Pain in Infants, Children, and Adolescents

Committee on Psychosocial Aspects of Child and Family Health

Task Force on Pain in Infants, Children, and Adolescents

PEDIATRICS 108 (3) 2001, 793 -797



**Procedural pain and distress**

The American Academy of Pediatrics (AAP) and the American Pain Society (APS) jointly issue this statement to underscore the responsibility of pediatricians to take a leadership and advocacy role to ensure humane and competent treatment of pain and suffering in all infants, children, and adolescents

“The concepts of pain and suffering go well beyond that of a simple sensory experience. It has emotional, cognitive, and behavioral components as well as developmental, environmental and sociocultural aspects”

“... It is important that Pediatricians expand their knowledge and advocate for the use of effective pain management in children”

**There is no ethical justification to withhold analgesia when a child is clearly in pain !**



**MYTHS AND MISCONCEPTIONS**  
**are still a major problem**

## **Myths that lead to undertreatment of pain**

- 1. “Young infants don’t feel pain as adults”**
- 2. “Young infants don’t remember pain”**
- 3. “Pain has no impact on brain development”**
- 3. “Pain is character building”**



## Young infants are more sensitive to pain than adults

Pain is a neuroadaptive phenomenon that emerges in the middle of pregnancy, at about 20–22 weeks of gestation

By 29 wks of gestation, pain pathways and cortical + sub-cortical centers involved in the perception of pain are well developed, as are the neurological systems for the transmission and modulation of pain sensation

Pain sensitivity in neonates is more profound than that of older individuals; their nervous system may be less effective at blocking painful stimuli than those of adults (immaturity of descending pain inhibitory pathways)



 "Young infants don't remember pain"

## The infant's CNS remembers pain

Young Infants who underwent painful procedures exhibited hyperalgesia (an increased response to a stimulus that is normally painful) during a subsequent procedure compared to normal infants who had not undergone repeated painful procedures.

The reduction of pain thresholds is a long-term effect of procedural pain on young infants (makes them more sensitive to pain stimuli)

Taddio A, Katz J, Ilersich AL, Koren G. Effect of neonatal circumcision on pain response during subsequent routine vaccination. *Lancet* 1997;349:599–603.

Taddio A, Shah V, Gilbert-MacLeod C, Katz J. Conditioning and hyperalgesia in newborns exposed to repeated heel lances. *JAMA* 2002;288:857–61

R. Slater, L. Fabrizi, A. Worley, J. Meek, S. Boyd, M. Fitzgerald. Premature infants display increased noxious-evoked neuronal activity in the brain compared to healthy age-matched term-born infants *Neuroimage*, 52 (2010), pp. 583-589

N.L. Maitre, A.P. Key, O.D. Chorna, J.C. Slaughter, P.J. Matusz, M.T. Wallace, et al. The dual nature of early-life experience on somatosensory processing in the human infant brain *Curr Biol*, 27 (2017), pp. 1-7

C. Hermann, J. Hohmeister, S. Demirakça, K. Zohsel, H. Flor. Long-term alteration of pain sensitivity in school-aged children with early pain experiences. *Pain*, 125 (2006), pp. 278-285

 "Pain has a negative impact on brain development"

## **Pain has a negative impact on brain development**

Early procedural pain in very preterm infants may contribute to impaired brain development.

Acute pain has long-term effect on behavior and cognitive ability.

Brummelte S, Grunau RE, Chau V, et al. Procedural pain and brain development in premature newborns. *Ann Neurol*. 2012 Mar;71(3):385-96.

J. Vinall, S.P. Miller, B.H. Bjornson, K.P.V. Fitzpatrick, K.J. Poskitt, R. Brant, et al. Invasive procedures in preterm children: brain and cognitive development at school age. *Pediatrics*, 133 (2014), pp. 412-421

“Pain is character building”

## **Pain is NOT character building**

This statement is unfair. It legitimizes pain and takes away the child's right for pain relief

Children younger than 8 years are not able to understand that short-term pain may have long-term benefit

Adolescents who had poorly managed pain procedures show an increased level of anxiety in subsequent pain situations

# **Misconceptions that lead to undertreatment of pain in children**

- 1. Lack of routine pain assessment in children**
- 2. Lack of knowledge regarding newer modalities and proper dosing strategies for the use of analgesics in children**
- 3. Fears of respiratory depression or other adverse effects of analgesic medications.**
- 4. The belief that preventing pain in children takes too much time and effort**



# **DISTRACTION AND SEDATION IN THE PEDIATRIC ED**

Procedural sedation and analgesia

## Children can easily reach a state of calm and even fall asleep



**Procedural sedation and analgesia**

# **Procedural Sedation and Analgesia in the Emergency Department**

American College of Emergency Physicians, 1998

“PSA is a technique of administering sedatives or dissociative agents with or without analgesics to induce a state that allows the patient to tolerate unpleasant procedures while maintaining cardiorespiratory function.”

# PSA medications in the ED:

## Route of administration

| Topical                                                                                               | Oral                                                         | IN/SL                                                                                                  | Rectal                                                                                                       | IM                                                                                                                     | IV                                                                                                                                            | INH                                                          |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| <div>EMLA</div> <div>Ametop Gel</div> <div>LAT</div> <div>Elamax</div>                                | <div>Midazolam</div> <div>Chloral hydrate</div>              | <div>Midazolam</div> <div>Fentanyl</div> <div>Dexmedtom.</div>                                         | <div>Midazolam</div> <div>Methohexital</div> <div>Pentob.</div> <div>Thiopental</div> <div>Chloral hy.</div> | <div>Midazolam</div> <div>Ketamine</div>                                                                               | <div>Midazolam</div> <div>Ketamine</div> <div>Fentanyl</div> <div>Propofol</div> <div>Etomidate</div> <div>Pentob</div> <div>Dexmetedo.</div> | <div>Nitrous Oxide</div>                                     |
|  <div>LAT gel</div> |                                                              |  <div>Atomizer</div> |                                                                                                              |                                                                                                                        |                                                                                                                                               |                                                              |
|                                                                                                       | <div>Flumazenil (Anexate)</div> <div>Naloxone (Narcan)</div> |                                                                                                        |                                                                                                              | <div><u>Remifentanyl</u></div> <div><u>Sufentanyl</u></div> <div><u>Alfentanyl</u></div> <div><u>Diamorphine</u></div> |                                                                                                                                               | <div>Morphine</div> <div>Oxycodone</div> <div>Tramadol</div> |

Procedural sedation and analgesia



The challenge lies in determining the optimal drug combination that will enable the procedure to be successfully completed

while simultaneously:

minimizing the patient's risk of adverse events

AND

promotes a smooth recovery

| <u>Indication</u>                            | <u>One Option</u>                                          | <u>Another Option</u>                                     |
|----------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------|
| <b>Laceration repair</b>                     |                                                            |                                                           |
| Face ≥ 6 y/o                                 | LAT <sup>a</sup> Gel and INH N <sub>2</sub> O <sup>b</sup> | IV Ketamine                                               |
| Face < 6 y/o                                 | LAT Gel and PO/IN Midazolam                                | IM Ketamine                                               |
| Finger ≥ 6 y/o                               | INH N <sub>2</sub> O and Digital nerve block <sup>c</sup>  | IV Ketamine and Propofol                                  |
| Finger < 6 y/o                               | PO/IN Midazolam and Digital nerve block                    | IV Ketamine and Propofol                                  |
| <b>Fracture reduction</b>                    |                                                            |                                                           |
| ≥ 6 y/o                                      | IV Ketamine and Propofol                                   | 1. IN Fentanyl and INH N <sub>2</sub> O<br>2. IV Ketamine |
| < 6 y/o                                      | IV Ketamine and Propofol                                   | IM Ketamine                                               |
| <b>Difficult IV access</b>                   |                                                            |                                                           |
| ≥ 6 y/o                                      | EMLA crème <sup>d</sup> ± INH N <sub>2</sub> O             |                                                           |
| < 6 y/o                                      | EMLA crème ± PO/IN Midazolam                               |                                                           |
| <b>Bladder Catheterization</b>               | PO/IN Midazolam                                            | INH N <sub>2</sub> O                                      |
| <b>Foreign body removal from ear or nose</b> | PO/IN Midazolam                                            | IV Ketamine and Propofol                                  |

<sup>a</sup> Topical mixture of lidocaine, adrenaline, and tetracaine.

<sup>b</sup> Recommended only in cooperative children.

<sup>c</sup> Warmed, buffered 1% lidocaine should slowly be injected using a 30-gauge needle.

<sup>d</sup> Should be placed on skin for at least 60 minutes before the procedure, on intact skin.

| <u>Indication</u>                            | <u>One Option</u>                                                   | <u>Another option</u>                         |
|----------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------|
| <b>Incision &amp; drainage of an abscess</b> |                                                                     |                                               |
| ≥ 6 y/o                                      | IV Ketamine                                                         | IV Ketamine and Propofol                      |
| < 6 y/o                                      | 1. IV Ketamine<br>2. IM ketamine                                    | IV Ketamine and Propofol                      |
| <b>Lumbar puncture</b>                       | EMLA crème and infiltration of local anesthetic and PO/IN Midazolam | 1. IV Ketamine<br>2. IV Ketamine and Propofol |
| <b>Debridement of a severe burn</b>          | 1. IN Fentanyl and IN midazolam<br>2. IN Ketamine                   | 1. IV Ketamine<br>2. IM ketamine              |
| <b>Arthrocentesis</b>                        |                                                                     |                                               |
| ≥ 6 y/o                                      | IV Ketamine                                                         | IV Ketamine and Propofol                      |
| < 6 y/o                                      | 1. IV Ketamine<br>2. IM ketamine                                    | IV Ketamine and Propofol                      |
| <b>Diagnostic imaging</b>                    | 1. IN Midazolam<br>1. IN Dexmedetomidine                            | IV Propofol                                   |

# THANK YOU

